# Packages in Java Demonstration

Title: Packages – Defining, Creating, Accessing and Importing Packages in Java

**Objective / Aim :**

To define a package named "number", create a "Roman" class inside it, implement the "romanToInteger()" method, and import the method in another class to convert a Roman numeral into an integer.

**System Requirements :**

Hardware:

- Standard PC/Laptop
- Intel Core i3/i5 processor or equivalent
- Minimum 4 GB RAM
- Minimum 500 MB free storage
- Keyboard and Monitor

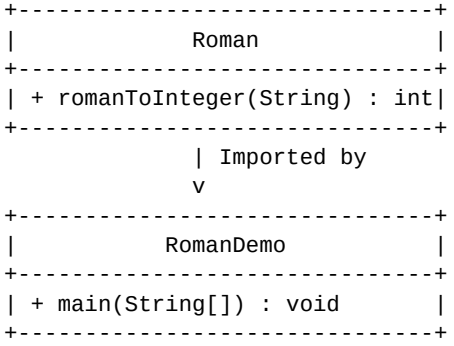
Software:

- Operating System: Windows/Linux/macOS
- Java Development Kit (JDK 17 or JDK 21)
- Visual Studio Code / Eclipse IDE or any Java IDE
- Command Prompt/Terminal

**Algorithm / Logic Description :**

1. Start the program.
2. Define a package named number.
3. Create a public class named Roman inside the number package.
4. Implement the romanToInteger(String s) method to parse each Roman numeral and compute its decimal integer value.
5. Import the number.Roman class into another class named RomanDemo.
6. Store the Roman numeral "LVIII" in a String variable.
7. Create an instance of Roman, call romanToInteger(), and display the result.
8. Stop the program.

**UML Class Diagram :**



Relationship: RomanDemo imports and accesses the public class Roman located in the user-defined package number.

**Source Code :**

File 1: Roman.java

```
package number;

public class Roman {

    public int romanToInteger(String s) {
        int result = 0;
        int previous = 0;

        for (int i = s.length() - 1; i >= 0; i--) {
            int value = 0;

            switch (s.charAt(i)) {
                case 'I': value = 1; break;
                case 'V': value = 5; break;
                case 'X': value = 10; break;
                case 'L': value = 50; break;
                case 'C': value = 100; break;
                case 'D': value = 500; break;
                case 'M': value = 1000; break;
            }

            if (value < previous) {
                result -= value;
            } else {
                result += value;
            }

            previous = value;
        }

        return result;
    }
}
```

File 2: RomanDemo.java

```
import number.Roman;

public class RomanDemo {
    public static void main(String[] args) {
        Roman obj = new Roman();

        String roman = "LVIII";
        int result = obj.romanToInteger(roman);

        System.out.println("Roman Numeral: " + roman);
        System.out.println("Integer Value: " + result);
    }
}
```

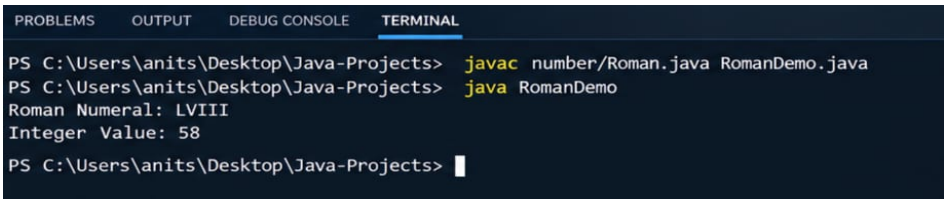
**Compilation & Execution Commands :**

Save files as: number/Roman.java and RomanDemo.java

Compilation: javac number/Roman.java RomanDemo.java

Execution: java RomanDemo

**Output Screenshot :**




**Result :**

Thus, the number package was created successfully, the Roman class was defined, and its romanToInteger() method was imported and accessed from another class to convert the Roman numeral "LVIII" into the integer "58".